COMPUTATIONAL ORDINAL SOCIOMETRY (NEXORD)

SOCIOMETRIA ORDINAL COMPUTACIONAL (NEXORD)

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«Toward a Physics of the Collective»

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ABSTRACT

Computational Ordinal Sociometry formally updates the foundational intuitions of Jacob Levy Moreno (Sociometry) and Kurt Lewin (Field Theory), overcoming the methodological constraints of their era through contemporary computation. This Manifesto articulates the model across seven foundational pillars:

1. **The conception of the relational bond as the constitutive core of the existential reality of groups.** «Giving and Receiving» form the indissociable poles of a continuous interactive circuit, where collective health, well-being, and viability depend on the homeostatic equilibrium between interpersonal exchanges of acceptance and conflict.
2. **The rigorous quantification of hierarchical choice order via the rank-inverse function**, formalizing that the human mind prioritizes and chooses ordinally without the need to impose artificial cardinal scales.
3. **The continuous metric arising from this quantification of preferential order**, formally articulated within the algebra of the relational tetragram and reciprocity metaperceptions, transforming interpersonal micro-interactions into a fully computable topology.
4. **The socio-thermodynamic modeling of the collective through density gradients (18D) and relational friction (Φ)**, wherein moderate dialectical discrepancy and internal plasticity serve as necessary conditions for learning and adaptation against the entropy of groupthink.
5. **The simultaneous projection across dual factorial planes**, where actors are concurrently represented as emitters and receivers of preferences in barycentric equilibrium, enabling the geometric comparison of group structures via topological manifolds.
6. **The versatility of the multidimensional design (GxTyCz)**, integrating cross-sectional, longitudinal, and plural task-and-affect criteria analyses, complemented by the interactive visualization tools of the VISORD platform.
7. **The governance commitment under Ethics by Design and Open Science principles**, guaranteeing an exclusively formative, non-punitive use of data aimed at safeguarding participant dignity and preserving empirical heritage through the

Sociometric Bank (**BSOC**).

Keywords: NEXORD Platform, VISORD, Sociometric Bank (BSOC), Computational Ordinal Sociometry, Preventive Sociometry, ORBIT Format, Rank-Inverse Function, Relational Densities (18D), Metaperception, Dual Factorial Planes, Grassmann Manifolds, Groupthink, Ostracism, Ethics by Design, Open Science.

1. Epistemological Foundations

«The human bond is not a mere numerical outcome; it is a NEXUS that only acquires meaning when it achieves ORDER. Without order, sociometry is reduced to a count of affinities; with order, sociometry becomes topology and thermodynamics of collective space.»

1.1. Overcoming the Classical Limit and the Ontology of the Relational Tie

The purpose of Computational Ordinal Sociometry in NEXORD is to methodologically update the foundational tradition of social psychology: translating into computational formalisms the insights that neither Jacob Levy Moreno nor Kurt Lewin could operationalize with the technical instrumentation of their era.

Moreno conceived the concept of the social atom and the deep architecture of affective preferences, but manual computation historically forced sociometric questionnaires into rigid quotas of binary choice and rejection (typically three positive choices and three rejections). Lewin formulated psychological field theory and hodological topology, but lacked continuous mathematical operators capable of quantifying psychological distances without fracturing the psychosocial nature of life space.

NEXORD positions active relational interaction at the ontological core of collective life: the group is not a mere aggregate of individual dispositional traits nor a static network of cold nodes, but a dynamic field of tensions, attractions, and resistances that continuously emerges and transforms.

As an explicit tribute to both founders, the NEXORD Platform commissioned two territorial maps in the tradition of Mademoiselle de Scudéry's Carte du Tendre (1654)—the cartographic-literary genre that inspired the name of the Clélie Protocol for data custody (§7.2)—: the Carte de la Spontanéité et du Télé for Jacob Levy Moreno (1934) and the Carte du Champ Dynamique et de l'Espace Vital for Kurt Lewin (1936–1951) (Figures 1 and 2).

These maps declare the epistemic status of the project: at no point does NEXORD seek to replace its predecessors; rather, it aims to complete their initial proposals, translating them into the computational language and capabilities of contemporary science.

Continuity with historical tradition is not merely cartographic: NEXORD operates interchangeably with MORENO and ORBIT capture protocols (closely related, dual-stack protocols, since both share manifest choice/rejection ordering; while ORBIT does not record

reciprocal expectations directly, a MORENO ↔ ORBIT conversion algorithm enables probabilistic estimation completing both sociometric forms). The dyadic tetragram (pDA DA RC pRC) is thoroughly Morenian in conception and forms the operational bedrock of the NEXORD Platform.

The specific debt to Kurt Lewin lies at the core of the motto «The Physics of the Collective»—and of the operationalization of its central variable, relational density, which anchors the socio-thermodynamic formal apparatus (§4.4)—rooted in field theory, which conceived group life as force dynamics within a measurable life space: what is quantified here as density is not decorative metaphor, but an algorithmic continuation of that original field intuition.

Figures 1 and 2. Foundational Cartographic Diptych: Carte de la Spontanéité et du Télé by Jacob L. Moreno (1934). Carte du Champ Dynamique & de l'Espace Vital by Kurt Lewin (1936–1951). Tribute by NEXORD.

1.2. From Qualitative Hierarchy to Continuous Field: The Inverse Rank Function

Without a standardized ordinal framework, classical sociometric records were difficult to compare. NEXORD provides a shared reference structure where sequential ranking allows group configurations to be systematically formalized, integrated, and compared (Table 1).

In the foundation of the scientific method, order conceptually precedes quantitative metrics. While assigning cardinal magnitudes to affective states assumes interval scales that do not always reflect the subjective decision process, preferential ranking aligns directly with the nature of relational judgment: individuals hierarchize, prioritize, and choose their ties.

Recording the sequential order of choice captures the structure of interpersonal preferences with fidelity without imposing a priori metric assumptions. In NEXORD, the continuous metric is not introduced as an external postulate, but emerges analytically from relational order through the inverse rank function.

The analytical mechanism enabling this transition from qualitative ordering to continuous calculus is the quantification of hierarchical order:

Wij = 1 / rijγ** (1) where rij ∈ 1, 2, ..., N represents the strict ordinal rank assigned by selector i to peer j, and γ modulates the interdistance between distinct ranks.

In its complementary bounded continuous formulation, normalized interpersonal density takes the form: **Zij = (N − rij) / (N − 1) ∈ [0, 1]** (2) where the top rank **(rij = 1)** reaches the unitary bound **(Z = 1.0)**, the bottom rank **(rij = N)** drops to absolute

zero (**Z = 0.0**), and intermediate ranks are distributed in strictly equispaced intervals.

It is essential to distinguish operator hierarchy: the inverse rank function (1) constitutes the fundamental operator for quantifying preference rank and drives the platform's relational engine (weights of the **SMIb** sociomatrices and field decay); the bounded linear transformation (2) is an operational simplification for metrics requiring strict bounds in [0, 1]. Both transformations enable the dyadic tie to be operated as a continuous and differentiable force, projectable onto complex geometric manifolds. This formal convergence aligns with modern statistical models for rank-order relational data (Krivitsky & Butts, 2017) and psychometric evidence regarding measurement degradation in truncated discrete quotas (Terry, 2000).

1.3. The Fundamental Axiom: GIVING and RECEIVING as an Indissociable Fact

Emitting preferences (GIVING) and receiving collective impact (RECEIVING) do not constitute independent psychological traits or isolated personality variables: they are polar, indissociable constituent moments of a single continuous relational reality.

The harmonious functioning or pathological fracture of a group does not reside in members taken in isolation, but in the fluidity, symmetry, and shear of their mutual exchanges. An actor may exhibit an intensely cooperative disposition while the group systematically relegates them to invisibility or rejection; divorcing these two poles leads to diagnoses blind to latent exclusion and to stigmatizing interventions.

2. ORBIT Format, Matrix Duality, and Diagonal Closure

In classical sociometry, the constraints of manual computation naturally favored tallying procedures based on fixed quotas (typically three choices and three rejections), where each nomination received equal unit weight (1 = 1), although it was soon replaced by a linear weighting inverse to rank (5–1).

2.1. Forced Drag without Ties and Matrix Duality A(i, k)=j vs. B(i, j)=k

The ORBIT Sociomatrix (initially termed the BREVE format due to its administrative speed) formalizes exhaustive preferential ordering **A(i, k) = j** over the complete collective, where the sender occupies their own position as an integral member. Total order allows representing a complete sequence of all group members. From this ordinal foundation, the model formalizes the matrix and continuous treatment of relational exchanges as a permutation of collective space.

This structure splits analytically into two complementary, bijective matrix structures:

- Direct Order Matrix A(i, k) = j: Records in each row i the peer j assigned to ordinal rank slot $k \in 1^\circ, 2^\circ, \ldots, N^\circ$.
- Distribution Sociomatrix **B(i, j) = k**: A square N × N sociomatrix where rows (i) and columns (j) indicate the rank order assigned: $B(i, j) = k \Leftrightarrow A(i, k) = j$.

Since each row constitutes a strict permutation of the set 1, 2, ..., N, the **ORBIT**

Sociomatrix invariably satisfies the property of row ordinal sum conservation:

$\sum_{j=1}^{N} B(i, j) = N(N+1)/2 = \text{const}$ (for $N = 6$, $\sum_{j=1}^{6} B(i, j) = 21$)

This algebraic invariance underpins the bidirectional, exact transposition between the classical Moreno sociomatrix and the ORBIT sociomatrix, as well as the subsequent derivation of relational tetragrams (Table 1):

- Classical Moreno matrix: Sociomatrix (E, R, pE, pR) in input format for N = 6: Transposition to ORBIT A(i, k)=j and B(i, j)=k.
- ORBIT sociomatrix: Format A(i, k)=j and B(i, j)=k, in Excel format for N = 6: Transposition to MORENO (E, R, pE, pR).
- SMI Matrices: Matrices SMIa(i, j) of the figurative tetragram and SMIb(i, j) of the alphanumeric tetragram.

MORENO					META		AM1(i, k) = j MORENO							BM1(i, j) = k MORENO						
IDi	E	R	pE	pR	αE	ωE	IDi	1º	2º	3º	4º	5º	6º	IDi	1A	2A	3A	4A	5A	6A
1A	3	4	3	4	3		1A	3	1				4	1A	2		1			4
2A	3, 1	4	3	4	3	1	2A	3	1		2		4	2A	2	4	1			4
3A	1, 2, 6	4	1	4	1	6	3A	3	1	2	6		4	3A	2	3	1	6		4
4A	3, 1	2	3	2	3	1	4A	3	1			2	4	4A	2	5	1			4
5A	3, 1, 6	4	3	4	3	6	5A	3	5	1	6		4	5A	3		1	6	2	4
6A	3, 1, 5	4	3	4	3	5	6A	3	6	1	5		4	6A	3		1	6	4	2

		META		Ab2(i,k)=j BREVE							Bb2(i,j)=k BREVE						
		αE	ωE	IDi	1º	2º	3º	4º	5º	6º	IDi	1A	2A	3A	4A	5A	6A
ORIGINAL	MORENO	3		1A	3	1	5	6	2	4	1A	2	5	1	6	3	4
		3	1	2A	3	1	5	2	6	4	2A	2	4	1	6	3	5
		1	6	3A	3	1	2	6	5	4	3A	2	3	1	6	5	4
TRANSPOSICION	BREVE	3	1	4A	3	1	5	6	2	4	4A	2	5	1	6	3	4
		3	6	5A	3	5	1	6	2	4	5A	3	5	1	6	2	4
		3	5	6A	3	6	1	5	2	4	6A	3	2	1	6	4	2

		ORIGINAL AB1(i,k)=j BREVE							BB1(i,j)=k BREVE						
		IDi	1º	2º	3º	4º	5º	6º	IDi	1A	2A	3A	4A	5A	6A
ORIGINAL	BREVE	1A	3	1	5	6	2	4	1A	2	5	1	6	3	4
		2A	3	1	5	2	6	4	2A	2	4	1	6	3	5
		3A	3	1	2	6	5	4	3A	2	3	1	6	5	4
TRANSPOSICION	MORENO	4A	3	1	5	6	2	4	4A	2	5	1	6	3	4
		5A	3	5	1	6	2	4	5A	3	5	1	6	2	4
		6A	3	6	1	5	2	4	6A	3	2	1	6	4	2

CONVERSION a MORENO					META		Am2(i, k) = j MORENO							Bm2(i, j) = k MORENO						
IDi	E	R	pE	pR	αE	ωE	IDi	1º	2º	3º	4º	5º	6º	IDi	1A	2A	3A	4A	5A	6A
1A	3, 1, 5	4, 2	3, 1	4	3		1A	3	1	5		2	4	1A	2	5	1	6	3	
2A	3, 1, 2	4, 6	3, 1	4	3	1	2A	3	1	2		6	4	2A	2	3	1	6		5
3A	3, 1, 2	4, 5	3, 1	4	1	6	3A	3	1	2		5	4	3A	2	3	1	6	5	
4A	3, 1, 5	4, 2	3, 1	2	3	1	4A	3	1	5		2	4	4A	2	5	1	6	3	
5A	3, 5, 1	4, 2	3, 5	4	3	6	5A	3	5	1		2	4	5A	3	5	1	6	2	
6A	3, 6, 1	4, 2	3, 6	4	3	5	6A	3	6	1		2	4	6A	3	5	1	6		2

Table 1. Comparative data input structure: top, classical Moreno questionnaire restricted to discrete choices/ rejections; bottom, the ORBIT Sociomatrix with exhaustive preferential ranking without ties satisfying invariably the ordinal sum conservation law.

The ORBIT format is particularly indicated for small groups ($N < 25$)—work teams, operational units, classrooms, or living groups—where exhaustive ranking is analytically and structurally optimal for capturing the relational microstructure with maximum resolution. For larger collectives ($N \geq 25$), NEXORD provides, when necessary, a successive-phase ranking procedure (a system extending Moreno's 3–5 quotas to broader and stratified

selections), without compromising metric resolution.

2.2. Reflexive Diagonal Closure and Core Sensors (IDA, ITA, P(k))

With both extremes of the list secured (E and R), the value assigned to intermediate positions is of negligible weight.

NEXORD resolves the secular omission of the sociometric main diagonal by requiring each participant to self-rank within the collective. This topological closure converts the matrix into a complete, continuous relational space, activating three fundamental metric sensors:

- Self-Rank Discrepancy Index (IDAi): Measures scalar divergence between projected self-image (rii) and external collective consensus received from the group (R_ext_i): IDAi = rii − R_ext_i. Negative values reflect hierarchical overestimation; positive values, peripheral self-inhibition or underestimation.
- Self-Rank Tension Index (ITA): Evaluates structural friction induced by leadership contestation when multiple participants claim top rank preference for themselves: ITA = max(0, ∑_i=1^N I(rii=1) − 1).
- Cognitive Partition and Combinatorial Anchoring P(k): By locating self-rank at position k, the subject's admissible combinatorial space is bounded to P(k) = ∩(N−1, k−1), formalizing that interpersonal preferences are emitted from a reference self-position: upward or downward relative to one's subjective horizon.

3. Relational Dynamics: Field Regimes, Friction, and Viability

3.1. Relational Friction Coefficient (Φ) and Structural Cohesion (ICE)

Internal resistance within the collective system is synthesized in the Relational Friction Coefficient (Φ) and its complementary metric, the Structural Cohesion Index (ICE):

$$\Phi = 2 / [N(N-1)] \sum_{i<j} |Z_{ij} - Z_{ji}|^{\alpha} \quad (\alpha \geq 1) \quad (3)$$

$$ICE = 1 - \Phi \quad (4)$$

Moderate values of Φ reflect internal dynamism and adaptive plasticity, whereas extreme values signal risks of fragmentation or destructive turbulence.

3.2. Psychologization of Field Regimes and Groupthink Prevention

Collective dynamics cannot be reduced to inanimate physics: system states express phases of psychological and interpersonal evolution:

- Rigidity and Structural Inertia: Rigid hierarchical crystallization, immobile roles, and scarce communicative permeability.
- Fluidity and Adaptive Cohesion: Open homeostasis with flexible circulation of preferences, agreements, and divergences.
- Dispersion and Anomie: Low coupling density, affective detachment, and absence of shared projects.
- Effervescence and Collective Fusion: High shared emotional activation and intense interactive mobilization.

From this perspective, NEXORD fundamentally challenges the traditional assumption that optimal group health equates to absolute cohesion (ICE = 1.0, Φ = 0). A collective devoid of discrepancies or internal friction is vulnerable to Groupthink (Janis, 1972), where pressures toward conformity suppress critical thinking and informed decision-making. Moderate friction (Φ ∈ [Φ_, Φ_]) and gradients across the GIVING/RECEIVING circuit provide the energy required for plasticity, organizational learning, and evolutionary adaptation.

3.3. The Living Puzzle Metaphor: Unjamming vs. Ecosystemic Relocation

Contrasting with traditional interventions that manage conflict through punitive exclusion or moral stigmatization of discordant members, NEXORD models the collective as a puzzle. An actor experiencing high friction is not an inherently pathological element: they typically suffer a decoupling of role expectations or an asymmetry of flows with their immediate dyadic environment.

Diagnostic facilitation promotes unjamming through task adjustments, communicative clarification, and opening channels of reciprocal collaboration. When, following systematic mediation, structural incompatibility remains persistent, the model envisions dignified ecosystemic relocation: migration to another group environment that preserves the psychosocial integrity of the individual and averts dynamics of wear or chronic ostracism.

3.4. The Ocean of Microstates and the Thermodynamic Stability Law

The combinatorial dimension of the sociometric sample space, even in a small group, is vast. For a group of five persons, each subject ordering four peers generates **4!** = 24 distinct orderings; combining the independent choices of the five members, total microstate space amounts to **24^5** = 7,962,624 matrices. If self-anchoring is included (**5!** = 120 orderings per row), the space surges to **120^5** ≈ 2.49 × 10^{10}—nearly 25 billion configurations—for a group of only 5 members. Faced with these numbers, thermodynamic stability and system health do not depend on chance, but on the geometry of energetic flow: minimizing shear dissipation and sustaining stable attractors.

The overwhelming majority of combinations correspond to thermal noise (dissonant matrices of high friction) or freezing traps (tectonic faults and suffocating monopolies). For a microstate to surpass thresholds of productive, healthy organization, it must simultaneously satisfy three thermodynamic invariants:

1 Minimization of shear dissipation: |Zij − Zji|^2 → 0 (reciprocity and mutual coherence).

2 Barometric gradient balance: (Ziin − Ziout)^2 ≤ ε (absence of black sinks or overloaded springs).

3 Global convective connectivity: existence of transitive return chains without isolated nodes (Ziin > Z_critical for all i).

Of this vast combinatorics, only a highly select percentage reaches an optimal relational state: group well-being is not the complete absence of tensions and conflicts, but an equilibrium among them, where no one is condemned to absolute zero nor forced to sustain alone the burden of the collective.

In synthesis: collective viability and well-being do not depend on individual voluntarism, but on the structural balance of mutual exchanges. A functional group sustains viable reciprocal flows, minimizes dissipative friction, and prevents extreme asymmetries where overload or isolation chronically concentrates on particular members.

4. Tensorial Algebra, Q81 Lattice, and Relational Density

4.1. 12-Symbol Atomic Grammar and the Relational Tetragram

The fundamental **Q81** lattice formalizes the complete combinatorial space of 3^4 = 81 dyadic configurations structuring the relational field. To provide the system with an exact, unambiguous notation, NEXORD establishes an alphabet of 12 symbols distributed across three relational polarity states:

SMIa Sintaxis

	DA		RECIBE		
tipo	**PERCEPCION**	**EFECTIVA**	**EFECTIVA**	**PERCEPCION**	**code**
elige	<	E	e	>	A-Z
indiferencia	¡	¿	?	!	0000
rechaza	[	R	r	]	a -z
	pDA	DA	RC	pRC	
			Tetragramas		
	<Ee>	¡Ee!	<¿r]	<E?>	....
	¡¿?!	<¿?]	¡E?>	[¿r!	....
	[Rr]	<Rr>	[¿?>	[Re!	....

Table 2. Syntax of the NEXORD relational alphabet: the 12 positional symbols

Each interpersonal interaction (i, j) is formalized as a relational figure composed of four elements or "tetragram" formed by the actions of GIVING and RECEIVING and two perspectives: expressed PREFERENCE and EXPECTATION regarding dyadic reciprocity **(p_DA, DA, RC, p_RC**). In turn, the elements of this tetragram or "relational unit" operate across three possible values (+, =, −). This structure configures a total of 3^4 = 81 relational archetypes, whose axis is defined by the 3 emblematic figures (Table 3):

IDi	1A	2A	3A	4A	5A	6A
1A	<Ee>	i¿e!	<Ee>	[Re!	i¿e!	i¿e!
2A	iE?!	<Ee>	<Ee!	[Rr]	i¿?!	i¿?!
3A	<Ee>	iEe>	<Ee>	[Re>	i¿e>	iEe>
4A	iEr]	[Rr]	<Er]	iRr!	i¿r]	i¿r]
5A	iE?!	i¿?!	<E?!	[R?!	<Ee>	iEe!
6A	iE?!	i¿?!	<Ee!	[R?!	iEe!	<Ee>

IDi	1A	2A	3A	4A	5A	6A
1A	BBBB	00E0	AAAA	ffF0	00C0	00D(
2A	0B00	DDDD	AAA0	ffff	0000	000(
3A	BBBB	0CCC	AAAA	ffFF	00EE	0DD(
4A	0Bbb	eeee	AAaa	0ff0	00cc	00d(
5A	0C00	0000	AA00	ff00	BBBB	0DD(
6A	0C00	0000	AAA0	ff00	0DD0	BBB(

Table 3. Transposition of sociometric data from MORENO and/or ORBIT formats to the figurative SMIa(i,j) and alphanumeric SMIb(i,j) matrices as fundamental analytical structures

The formalization of signed relational configurations (+, −) and the measurement of dyadic friction dialog constructively with contemporary research on signed graphs, communicability geometry, and structural frustration indices (Aref & Wilson, 2021; Cucuringu et al., 2023; Diaz-Diaz & Estrada, 2025).

4.2. Antidiagonal Topology of the (9 × 9) Lattice and [1, 81] Numeralization

The 9 × 9 partition crosses the 9 active interaction profiles
(rows i = **<E, ¡E, [E, <¿, ¡¿, [¿, <R, ¡R, [R**)
with the 9 perception profiles
(columns j = **e>, e!, e], ?>, ?!, ?], r>, r!, r]**).

The numeralization of the 81 relational figures (**Q81(9, 9)**) is resolved geometrically through the ordered sweep of the 17 secondary antidiagonals, traversed from bottom to top: each antidiagonal defines an isoline of constant net relational energy and valence, such that all figures located on the same antidiagonal share identical polarity sums between actual actions and cognitive expectations.

This geometric ordering provides the primary metric-scalar quantification of dyadic relational density (DQ81 ∈ [1, 81]), structured into three fundamental regions:

- Dissonance and Rejection Vertex (Scalars 1 to 36): The first 8 lower antidiagonals cover ∑_k=1^8 k = 8 × 9/2 = 36 figures located in the zone of conflict, wear, and relational retraction, starting from the pole of maximum negative density: [Rr] = 1.
- Neutral Baricentric Isoline (Scalars 37 to 45): The central transversal antidiagonal (k = 9, consisting of 9 cells) spans scalars 37 to 45. Its central cell (coordinate 5, 5) unequivocally fixes baseline indifference: Cell(5, 5) i¿?! = 41.
- Cohesion and Resonance Vertex (Scalars 46 to 81): The 8 upper antidiagonals cover the 36 figures of attraction, convergence, and synergy up to the apex of antidiagonal 17 (length 1), crowning the space in the figure of full harmonic resonance: <Ee> = 81.

Continuous Normalization of Q81 Density:

By centering the scale on the neutral indifference centroid (41), the elementary dyadic density of the figure is normalized in the standard continuous range [-1.00, +1.00]:

D_Q81(i, j) = Q_81(i, j) − 41 (13) where: DQ81(<Ee>) = Maximum Positive Density (+1), DQ81(i¿?!) = Neutral Density (0), and DQ81([Rr]) = Maximum Negative Density (−1).

This geometry preserves strict antipodal complementary symmetry: Antipode(k) = 82 − k, implying that Figure(i, j) + Figure(10−i, 10−j) = 82, demonstrating that the neutral centroid is self-invariant under polar inversion (41 + 41 = 82).

Q81(9,9)	NUMERALIZACION de FIGURAS RELACIONALES SOCIOMETRICAS								
FIGURAS	e>	e!	e]	?>	?!	?]	r>	r!	r]
<E	81 <Ee>	80 <Ee!	78 <Ee]	75 <E?>	71 <E?!	66 <E?]	60 <Er>	53 <Er!	45 <Er]
¡E	79 ¡Ee>	77 ¡Ee!	74 ¡Ee]	70 ¡E?>	65 ¡E?!	59 ¡E?]	52 ¡Er>	44 ¡Er!	36 ¡Er]
[E	76 [Ee>	73 [Ee!	69 [Ee]	64 [E?>	58 [E?!	51 [E?]	43 [Er>	35 [Er!	28 [Er]
<¿	72 <¿e>	68 <¿e!	63 <¿e]	57 <¿?>	50 <¿?!	42 <¿?]	34 <¿r>	27 <¿r!	21 <¿r]
¡¿	67 ¡¿e>	62 ¡¿e!	56 ¡¿e]	49 ¡¿?>	41 ¡¿?!	33 ¡¿?]	26 ¡¿r>	20 ¡¿r!	15 ¡¿r]
[¿	61 [¿e>	55 [¿e!	48 [¿e]	40 [¿?>	32 [¿?!	25 [¿?]	19 [¿r>	14 [¿r!	**10 [¿r]**
<R	54 <Re>	47 <Re!	39 <Re]	31 <R?>	24 <R?!	18 <R?]	13 <Rr>	9 <Rr!	**6 <Rr]**
¡R	46 ¡Re>	38 ¡Re!	30 ¡Re]	23 ¡R?>	17 ¡R?!	12 ¡R?]	8 ¡Rr>	5 ¡Rr!	**3 ¡Rr]**
[R	37 [Re>	29 [Re!	22 [Re]	16 [R?>	11 [R?!	7 [R?]	4 [Rr>	2 [Rr!	**1 [Rr]**

Table 4. Q81(9,9) Lattice: classification of the 81 possible dyadic relational figures, indexed by antidiagonals of decreasing valence (81=<Ee>, 41=¡¿?! and 1=[Rr]).

4.3. The Subject as a Structural Unit

The premise of attributing to each group member a single relational figure and a single overall density is viable only if a mere indiscriminate average of all dyadic ties is discarded. NEXORD preserves topological bidimensionality by splitting the subject i's position through their Structural Triptych:

- *Emission (DA): Fi^out = numerical Q81 average of emitted choices and expectations (i, j), determining the emitting semi-tetragram Fi^out = 'xx..'.*
- *Reception (RC): Fi^in = numerical Q81 average of received choices and expectations (j, i), determining the receptive semi-tetragram Fi^in = '..yy'.*
- *Global Figure of the Subject: From the joint articulation of both poles emerges their integral tetragram Fi^global = 'XXYY', whose Q81 value corresponds to the resulting figure.*
- *Self-Evaluation (AUTO): The self-positioning tetragram Fi^auto = 'zzzz' (the reflexive position attributed by the individual to themselves).*

4.4. The 18D Relational Hyperspace and the 3 Dual Prisms

4.4.1. Dyadic Modulators and Architecture of the 18 Relational Densities

Dyadic relational density quantifies the thermal and structural mass of each interpersonal tie across the thermodynamic continuum (ℝ). To ensure maximum reproducibility and analytical rigor, the procedure processes each interaction (i → j) through six objective mathematical modulators (Table 5, Algorithmic Traceability Matrix)

DENSIDAD RELACIONAL = W * M * F *R * P		
W= W1 + W2 + W3 + W4	M = Mrec * Mclar * Mesf	F *R * P
W1= (< = 2.2), (¡ =1), ([=0.4)	Mreciprocidad [0-5, 1, 1.5]	F = Q81:num/41
W2= (E = 1.9, (¿ =1), (R =0.8)	Mclarividencia [0-5, 1, 1.5]	R = 1/x**k
W3= (e = 2.4), (? =1), (r=0.6)	Mesfuerzo [0-5, 1, 1.5]	P= 0 .5 + ((Si -Smin) / (Smax - Smin))
W4= (> = 3.5), (! =1), (]=0.2)		

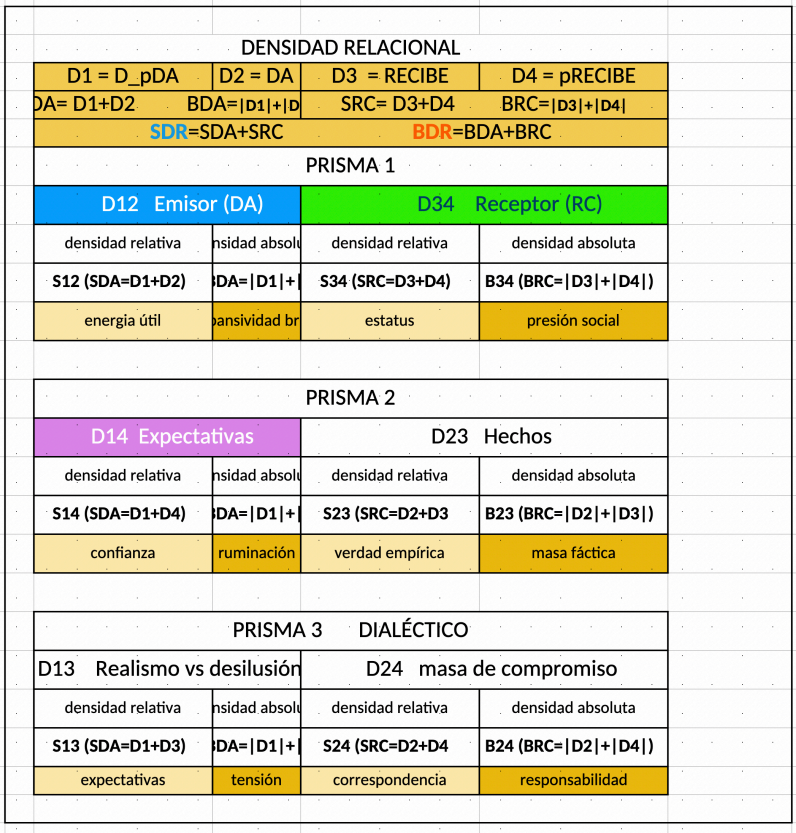
Table 5. Algorithmic Traceability Matrix and Dyadic Density Modulators

W: Establishes the initial bond potential between tetragram elements..
M: Modulators: 1) Dyadic Reciprocity, Action congruence (Ee, Rr, ¿?)
2) Clairvoyance, concordance between expectation and actual choice (<e, ¡?, [r)
3) Dyadic rank-concordance effort
F: Relational figure value in Q(81)
R: Inverse rank function
P: Sender prestige, sender network centrality
O: Opposition of Sentiments, dissonances and paranoias ((], Er, [>) $-(Pi \times 0.5)$
Negative weighting proportional to sender centrality:

The intersection between emission and reception in the sociomatrix generates the 18-density relational space (18D), governed by two prefixes: S (relative balance with sign $\mathbb{R}$, vector of attraction or rejection) and B (modular gross density $|\bullet| \geq 0$, total mass and mobilized investment).

These 18 densities are structured analytically across three orthogonal axes (Table 6):

- Prism 1 — Operational (10D): Decouples active giving from receptive status.
- Prism 2 — Ontological (14D): Contrasts the cognitive plane of expectations (SP, BP) with the factual plane of real choices and rejections (SA, BA).
- Prism 3 — Dialectical (18D): Evaluates relational realism (S13, B13, confirmation vs. disappointment) and dyadic correspondence or commitment (S24, B24).
- Global Invariants: Total Relative Density Balance (SDR) as a homeostatic barometer of relational climate, and Gross Relational Density (BDR) as an indicator of total group energy and mass.



DENSIDAD RELACIONAL			
D1 = D_pDA	D2 = DA	D3 = RECIBE	D4 = pRECIBE
DA= D1+D2	BDA=|D1|+|D	SRC= D3+D4	BRC=|D3|+|D4|
SDR=SDA+SRC		BDR=BDA+BRC	

PRISMA 1			
D12 Emisor (DA)		D34 Receptor (RC)	
densidad relativa	nsidad absolu	densidad relativa	densidad absoluta
S12 (SDA=D1+D2)	DA=|D1|+|	S34 (SRC=D3+D4)	B34 (BRC=|D3|+|D4|)
energia útil	ansividad br	estatus	presión social

PRISMA 2			
D14 Expectativas		D23 Hechos	
densidad relativa	nsidad absolu	densidad relativa	densidad absoluta
S14 (SDA=D1+D4)	DA=|D1|+|	S23 (SRC=D2+D3	B23 (BRC=|D2|+|D3|)
confianza	ruminación	verdad empírica	masa fáctica

PRISMA 3 DIALÉCTICO			
D13 Realismo vs desilusión		D24 masa de compromiso	
densidad relativa	nsidad absolu	densidad relativa	densidad absoluta
S13 (SDA=D1+D3)	DA=|D1|+|	S24 (SRC=D2+D4	B24 (BRC=|D2|+|D4|)
expectativas	tensión	correspondencia	responsabilidad

Table 6. 18D Relational Density Space with 3 Dual Prisms (operational {12-34}, ontological {14-23}, and dialectical {13-24}), derived from the elemental tetragram (D1 to D4), in gross densities (wear and decay) and relative balances (constructive energies), with invariant global balances (SDR, BDR

4.4.2. The 3 Dual Prisms and the 18D Pentagram

The four elemental densities D1, D2, D3, D4 admit three independent orthogonal dual partitions or Prisms (Figure 3): Operational (10D: Giver vs. Receiver), Ontological (14D: Mind vs. Facts), and Dialectical (18D: Realism vs. Commitment).

Architecture of the 18D Relational Space and the 3 Dual Prisms: Structuring of the tetragram into 4 elemental densities (D1–D4), global invariant balances (SDR, BDR), and decomposition across three fundamental axes: Operational (10D), Ontological (14D), and Dialectical (18D).

In the operational plane, the Socio-Thermodynamic Pentagram (Figure 3) articulates the ten primary densities by decoupling active emission (SDA, BDA) from receptive status (SRC, BRC), culminating in global invariants SDR and BDR. The analytical discrepancy between total mobilized energy and resulting net flow formally defines Relational Wear Inertia (I_wear = BDR − SDR), quantifying energy dissipated in friction and structural discrepancy before open fractures emerge in the group.

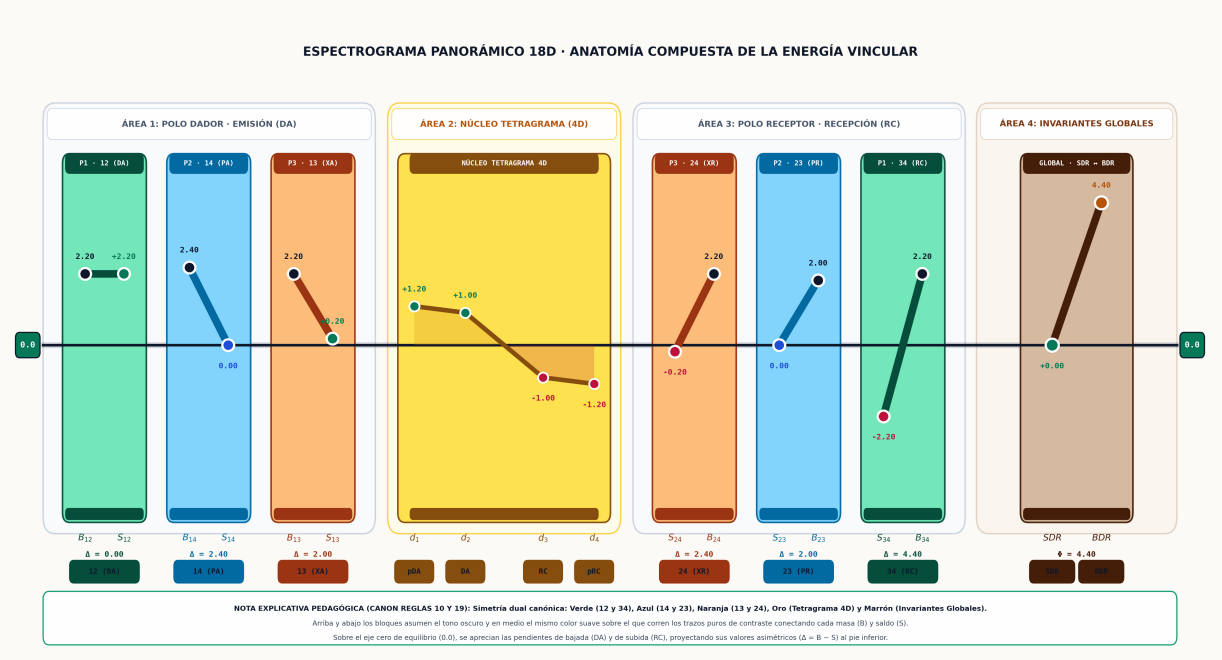


Figure 3. Panoramic 18D Spectrogram: B12 S12 (DA) B14 S14 (PA) B13 S13 (XA) [D1 D2 D3 D4 (TETRA)] S24 B24 (XR) S23 B23 (PR) S34 B34 (RC) SDR BDR. The shaded band between each gross mass (B) and its relative balance (S) quantifies the thermal friction gap (Φ = B – S).

4.5. The VISORD Aquarium Paradigm and Relational Hydrostatics

As the perceptual and experimental culmination of this socio-thermodynamic formalism, the VISORD Aquarium environment provides a complementary NEXORD application facilitating integrated visual and auditory perception of relational density structures within a multidimensional GxTyCz design (Figure 3b).

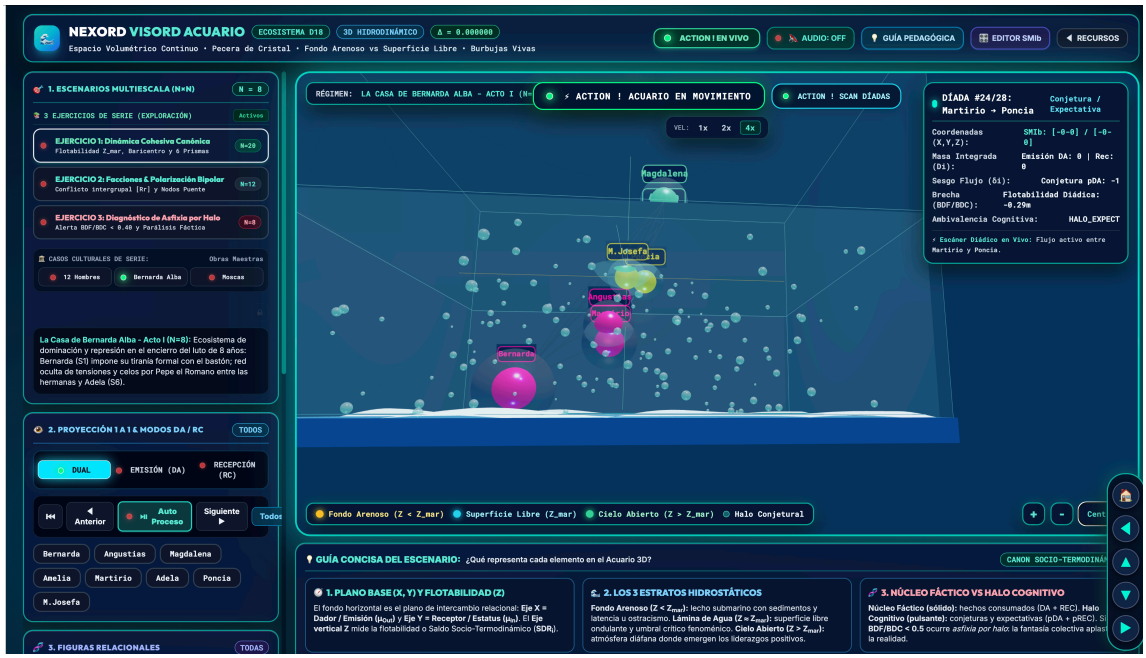


Figure 3b. *Hydrostatic Three-Dimensional Interface of VISORD Aquarium: Continuous Volumetric Space and Relational Buoyancy Dynamics (Cultural Case: The House of Bernarda Alba, Act I, N = 8)*

The scenario is modeled as a continuous three-dimensional space defined over the Cartesian volume DA × RECEIVE × DENSITY, analytically structured into three characteristic zones and strata:

- Lower stratum: Concentration of latent tensions, mutual rejection ([Rr]), and tie disconnection.
- Middle / neutral plane: Baseline level of social buoyancy corresponding to the threshold of indifference (i¿?! = 41).
- Upper stratum: Region of high positive density, harmonic cohesion (<Ee>), and integrative leadership with low friction.

In this scenario, different relational density values are projected in complete disjunction:

- Intra-subject level: Differential projection of the distinct relational pairs and kinematic dipoles of a single actor relative to their peers.• • Unit level: Distribution of subjects as dual agents: active Givers of preference (DA) and Receivers of group impact (RECEIVE).
- Macro-relational level: Aggregated global densities across Groups (G), Times (T), and Criteria (C).

5. Graphical Solutions, Multidimensional Space GxTyCz

5.1. Graphical Solutions: Visual Representation and Relational Topography

VISORD emerged from the necessity to render relational processes and structures resulting from investigation directly perceptible. The human eye and ear are ill-equipped to assimilate streams of numbers and abstract coefficients; conversely, they possess an extraordinary capacity to intuitively perceive shapes, reliefs, tonal shifts, relative distances, densities, and lighting.

The mathematical formalization of Ordinal Sociometry in NEXORD has advanced, through the tools of the VISORD environment, in developing visual and interactive representations that facilitate a deeper understanding of group interaction phenomena. As contemporary scientific development attests, mathematical parameterization renders perceptible dynamics that escape direct intuitive observation. This progress has been made possible through preference order quantification via the inverse rank function.

NEXORD articulates a suite of continuous graphical solutions designed to restore the relational field without perceptual saturation:

- Parallel Dual Factorial Planes (Biplot MESO F1 × F2, dual space CA ∥ PCA): Simultaneous projection where the dyadic vector DA → RC structures the position of each subject with barycenter at the origin of coordinates. In CA, across the contingency table of discrete integer counts; in PCA, across the standardized continuous density matrix. The contrast evidences the substantive contribution of relational socio-thermodynamics (Figure 4).

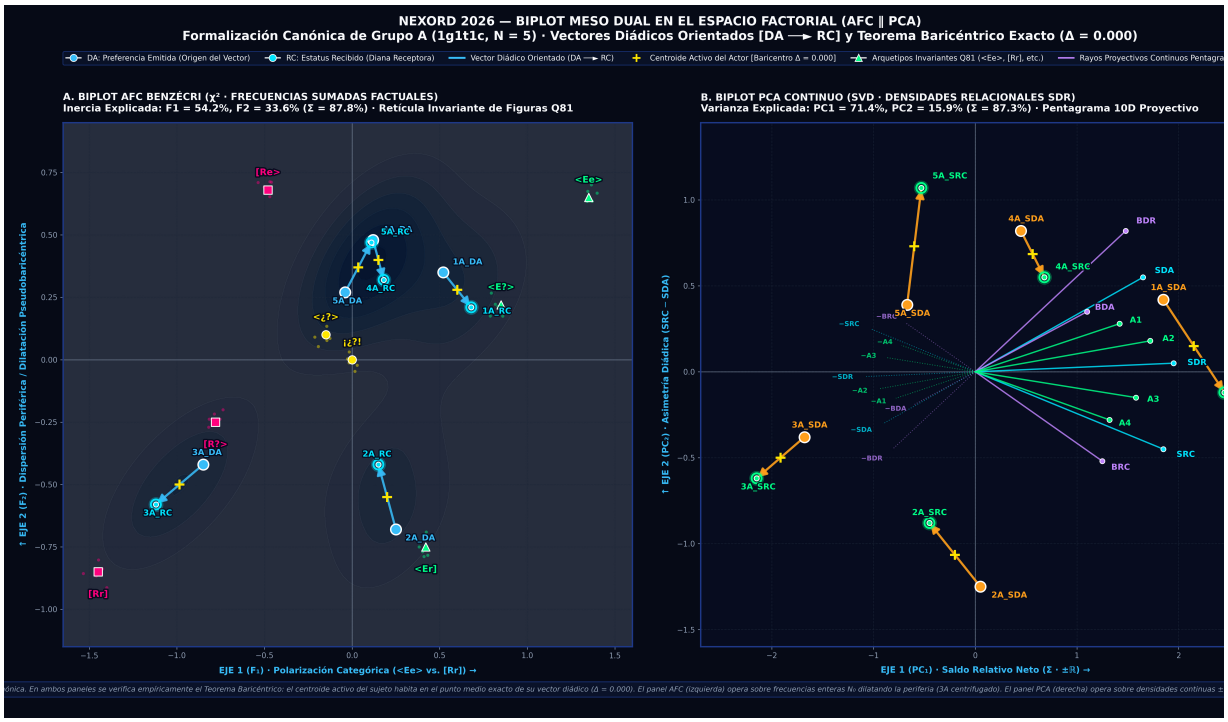


Figure 4. Parallel dual planes (Biplot MESO F1-F2, dual space CA-PCA): the dyadic vector points from the sender pole (DA) toward the receiver (RC), with barycenter at the origin.

- 1D Slender Vertical Factorial Axes: Projections derived from spectral decomposition and correspondence analysis (see Annex II).
- Three-Dimensional Continuous Sociograms (VISORD 3D/4D): Visualization in continuous space of attraction regions (<Ee>), intermediate opposition zones, and breakdown or friction filaments ([Rr]), with diachronic temporal navigation (see Annex I).
- Hierarchical Axonometric Projection onto a 2D plane of up to 6 dimensions (Author's DIMO Algorithm): To overcome classical 2D plane limitations without losing intuitive Euclidean interpretability, NEXORD incorporates in its algorithm the hierarchical axonometric projection method of up to 6 factorial dimensions developed by the author.
- Intergroup Structurograms: Relational support polygons for comparative morphological analysis across collectives.

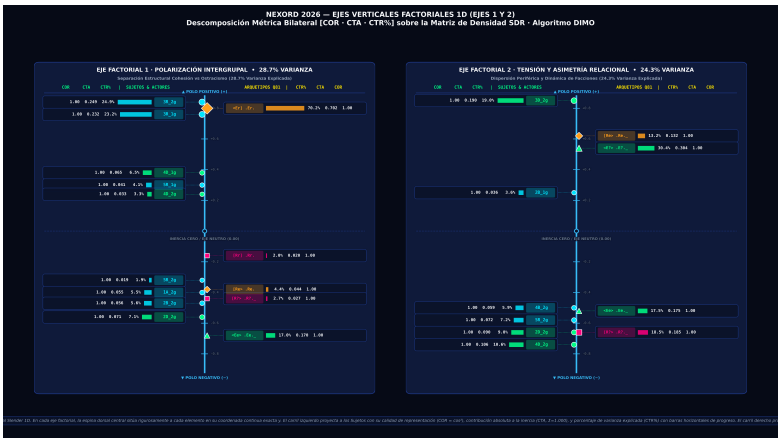


Figure 5. 1D Factorial vertical axes (Axes 1 and 2, SDR matrix): bilateral metric quality and 2D saturation resolution via 6D axonometric projection.

5.2. Multidimensional Space GxTyCzNEXORD is configured as a Psychosocial Sociometric Research Platform that transcends traditional sociometric testing as an isolated, static, and unidimensional snapshot. The platform conceives the field of human interactions as an open, continuous, and parameterizable tensorial research space, formalized over the Cartesian product G × T × C where researchers can design everything from single sociomatrix studies to ambitious longitudinal and intergroup comparative programs. Matricial formulation is articulated across four fundamental axes:

1 Axis G — Collectives and Groups (G = G1, G2, ..., Gx):

Structural transversality. The system accommodates operation on a single natural collective as well as constellations of independent groups, departments within a single organization, classroom networks, or operational units in critical missions. This axis enables comparative intergroup morphology (via geodesic distances on Grassmann manifolds) without forcing artificial cross-interactions between members belonging to distinct living environments.

1 Axis T — Longitudinal Temporal Waves (T = T1, T2, ..., Ty):

Longitudinal investigation, where sociometric tracking of groups over time enables evaluating their evolution and contrasting the effects of situations, critical incidents, or strategic interventions. NEXORD monitors diachronic cohesion evolution, status sedimentation velocity, leadership inertia or mutability, encapsulation or dissolution of latent tensions ([Rr]), and kinematic trajectories in density space. This longitudinal perspective enables diagnosing whether the collective evolves toward adaptive fluidity or collapses into rigidity, polarization, or wear traps before open fractures occur.

1 Axis C — Plural Sociometric Criteria (C = C1, C2, ..., Cz):

Diversity of evaluative criteria. Preferences and expectations in dyadic relations are not static; they vary substantially depending on evaluative criteria employed: operational, motivational, cognitive, socioaffective, leadership, etc.:

- Instrumental or task criteria (C1): technical competence, perceived efficacy, operational coordination, and workload distribution.
- Socioaffective or expressive criteria (C2): sympathy, emotional affinity, interpersonal trust, and informal coexistence.
- Stress or crisis leadership criteria (C3): reliability under pressure, conflict arbitration, and authority delegation under uncertainty.

The platform calculates inter-criteria discrepancy matrices and ordinal correlation coefficients revealing fractures invisible to classical analysis: individuals highly sought for tasks but affectively isolated, or informal leaders with strong affective appeal but disconnected from operational objectives.

1 Axis Z — Complementary Contextual and Psychosocial Variables (Z):

The analytical bridge between relational topology and individual attributes. Although intrinsic sociometric computation operates strictly on pure rank preferences (tie structure), NEXORD integrates two standard calibration batteries:

- Sociodemographic Battery V10: ten contextual variables (in 3 homogeneous modalities per item) locating actors along biographical, formal role, tenure, or institutional status coordinates.
- Group Adjectival Attribution Battery A16: sixteen adjectives grouped into 4 factors (thematic, functional, cognitive, affective) (GAA / AAG; Vicente, Cornejo & Barbero, 2006), evaluated on an even scale (1–8, without neutral midpoint) to map the cognitive-emotional climate and evaluative atmosphere projected onto the group.

Projected as illustrative or supplementary variables onto dual factorial planes (CA ‖ PCA), Z variables enrich diagnostic interpretation without altering the metric purity of the relational field. Furthermore, maintaining this common standard underpins the Sociometric Bank (BSOC) reference framework ensuring historical and retrospective comparability of sociomatrices across eras, cultures, and organizational settings.

The A16 battery is structured into four main factors grouping sixteen normative adjectives:

- Thematic (TEMA): clear, coherent, flexible, consensual.
- Functional (FUNC): productive, orderly, active, exemplary.
- Cognitive (COGN): open, critical, perceptive, expressive.
- Affective (AFEC): friendly, loyal, collaborative, tolerant.

For each adjective, dichotomic division of group evaluation allows detecting, via density centroids at both extremes, distinctive sensitivities in the perception of Group Activity, crossed with sociodemographic variables.

VARIABLES COMPLEMENTARIAS						
VAR	variable	mod	1	2	3	
SX	Género	3	Hombre	Mujer	HM	
ED	Edad	3	Joven	Adulta	Senior	
ORIG	Origen	3	Local	Pais	Foráneo	
ANT	Antigüed	3	Nuevo	Socio	Veterano	
ORIEN	Orientaci	3	Técnica	Social	Artística	
IMPL	Implicació	3	Baja	Media	Alta	
PRES	Presencia	3	Remota	Presencia	Híbrida	
ROL	Rol	3	Base	Medio	Mando	
REND	Rendimie	3	Bajo	Medio	Alto	
COM	Comunica	3	Deficiente	Regular	Optimo	
FACTORES ANÁLISIS DE LA ACTIVIDAD GRUPAL (AAG)						
TEMA	Temático	4	claro	coherente	flexible	consensuado
FUNC	Funcional	4	productivo	ordenado	activo	ejemplar
COGN	Cognitivo	4	abierto	crítico	perceptivo	expresivo
AFEC	Afectivo	4	amistoso	leal	colaborador	tolerante

Table 7. Battery of 10 contextual variables (V10, 3 modalities) and 16 group attribution adjectives (A16, 4 × 4, even scale 1–8 without neutral midpoint).

Applied deployment of the GxTyCz space is channeled through 10 ecosystemic calibration domains (Table 8), fixing structural field parameters—relational inertia α and decay rate γ—along with psychosocial focus and standard choice criteria across institutional application contexts (educational, family, sports, community, critical missions, therapeutic, corporate, institutional, military, and correctional).

10 Dominios Sociométricos Ecosistémicos			
Dominio Sociométrico	**Parámetros Estructurales**	**Foco Psicosocial y Dinámica Relacional**	**Criterios Clave de Elección**
1. Sociometría Educativa y Formativa	**α = 0.50 γ = 0.80**	Aprendizaje colaborativo, inclusión y prevención activa de la marginación escolar.	• Tarea / Estudio • Lúdico / Patio • Liderazgo Integrador
2. Sociometría Familiar y Sistémica	**α = 0.85 γ = 0.99**	Jerarquía, fronteras de rol, lealtades implícitas y resolución de tabúes sistémicos.	• Confianza y Sostén • Fricción / Conflicto • Toma de Decisiones
3. Sociometría Deportiva y Alto Rendimiento	**α = 0.30 γ = 0.60**	Sinergia táctica bajo presión, cohesión de vestuario vs. eficacia en campo.	• Pase / Sinergia Táctica • Cohesión de Vestuario • Liderazgo bajo Estrés
4. Sociometría Social y Comunitaria	**α = 0.30 γ = 0.40**	Dinámicas informales de barrio, agrupaciones espontáneas e integración vecinal.	• Proximidad y Afecto • Organización e Iniciativa • Compartir Ocio
5. Sociometría de Equipos de Élite / Misiones Críticas	**α = 0.85 γ = 0.90**	Fiabilidad absoluta, respuesta coordinada ante imprevistos y gestión de fallos.	• Fiabilidad Técnica • Respuesta a Crisis • Apoyo en Misión
6. Sociometría Terapéutica y Clínica	**α = 0.40 γ = 0.70**	Salud de la red de apoyo, vulnerabilidad, sostén emocional y contención afectiva.	• Sostén Emocional • Identificación y Reflejo • Distancia Defensiva
7. Sociometría Laboral y Organizacional	**α = 0.65 γ = 0.85**	Red informal de trabajo, eliminación de silos, clima laboral y retención de talento.	• Resolución Operativa • Innovación y Confianza • Fluidez Informativa
8. Sociometría Institucional y Gubernamental	**α = 0.80 γ = 0.90**	Coordinación interdepartamental, estabilidad normativa y consensos de gestión.	• Protocolo Institucional • Consenso Normativo • Consulta Estratégica
9. Sociometría Militar y Geo-Política	**α = 0.90 γ = 0.95**	Estructuras de mando, bloques diplomáticos, alianzas de poder y dinámicas de veto.	• Cadena de Mando • Lealtad de Bloque • Veto y Tensión
10. Sociometría Biológica / Eco-Penitenciaria	**α = 0.20 γ = 0.50**	Autoorganización, asignación de recursos escasos, nichos territoriales y supervivencia.	• Proximidad Espacial • Defensa y Cooperación • Dominancia Ordinal

Table 8. The 10 ecosystemic application domains with structural parameterization (α, γ), psychosocial focus, and choice criteria.

5.3. Multigroup PCA and Geodesic Distance on Grassmann Manifolds Gr(k, N)

To compare collectives without forcing artificial cross-interactions, each group is projected as a k-dimensional subspace within $\mathbb{R}^N$. From Jordan principal angles $(\theta_m) = \sigma_m(U1^T U2)$, scale-invariant geodesic distance is computed: $$\mathbf{d_{Gr}(U_1, U_2) = \sqrt{[\sum \theta m^2]}} \quad (20)$$

This metric rigorously determines (Figure 6) whether two groups share identical relational morphology regardless of sample size differences. While inter-rater agreement is

an appreciable component of reliability for invariant structure findings, it is the temporal persistence and constancy of these configurations in geometric space that allows testing the robustness of relational structures against conjunctural variations or random fluctuations.

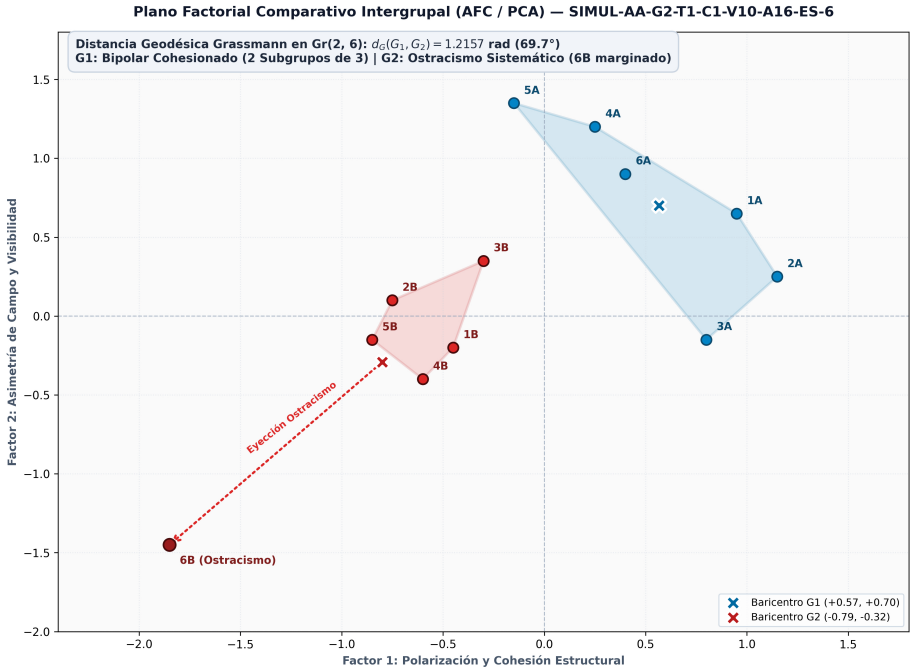


Figure 6. Comparative intergroup factorial plane (G1 vs G2) and diachronic structurograms; geodesic distance on Gr(k, N) quantifies shared relational morphology.

At the individual level, the Euclidean rank interdistance matrix dDA / dRC (Table 9) (Figure 7) documents over the G2T1C1 corpus the law of relational asymmetry: mean receiver dispersion (4.84; SD = 0.64) systematically exceeds sender dispersion (4.15; SD = 0.49), and positional diagnoses per node—from the isometric consonance of 5A (Δ = −0.09) to the maximum imbalance of 3A (Δ = −1.39)—illustrate the granularity of socio-thermodynamic diagnosis.

6. Experimental Research

6.1. Experimental Monitoring in the Physical Laboratory

The platform is designed for monitoring complex experimental investigations in in-person laboratory settings: diachronic tracking of highly interdependent tasks, crisis simulations, and continuous evaluation of interventions in small-group dynamics.

6.2. Dialectical Sociometry and Social Dilemma Analysis

In situations of cooperation and conflict (e.g., common-pool resource dilemmas or iterated prisoner's dilemmas), NEXORD quantifies the decoupling between cognitive expectations and observable relational behavior, capturing the emergence of alliances, defections, and reciprocity norms. Administering the ORBIT format to a group as PRE and/or POST in experimental designs proves to be an effective strategy for contrasting dependent and independent variables.

6.3. Multimodal Perception: Hydrodynamic Sonification

NEXORD implements the iceberg model along two complementary fronts:

- 3D Graphical Representation: Brings to the visual surface the submerged structure of tensions, latent rejections, and informal ties that remain unexpressed in the formal hierarchy.

INTERDISTANCIAS EN EL GRUPO						
	1A	2A	3A	4A	5A	6A
1A	0.00	0.85	1.16	2.22	2.22	2.42
2A	0.85	0.00	1.05	2.24	2.30	2.33
3A	[1.16	1.05	0.00	2.43	2.37	2.19
4A	2.22	2.24	2.43	0.00	0.73	1.19
5A	2.22	2.30	2.37	0.73	0.00	1.05
6A	2.42	2.33	2.19	1.19	1.05	0.00

Table 9. Numerical matrix of Euclidean rank interdistances (d_DA, d_RC, Δ = d_DA − d_RC) with structural diagnosis per subject, corpus G2T1C1: empirical validation of the relational asymmetry law.

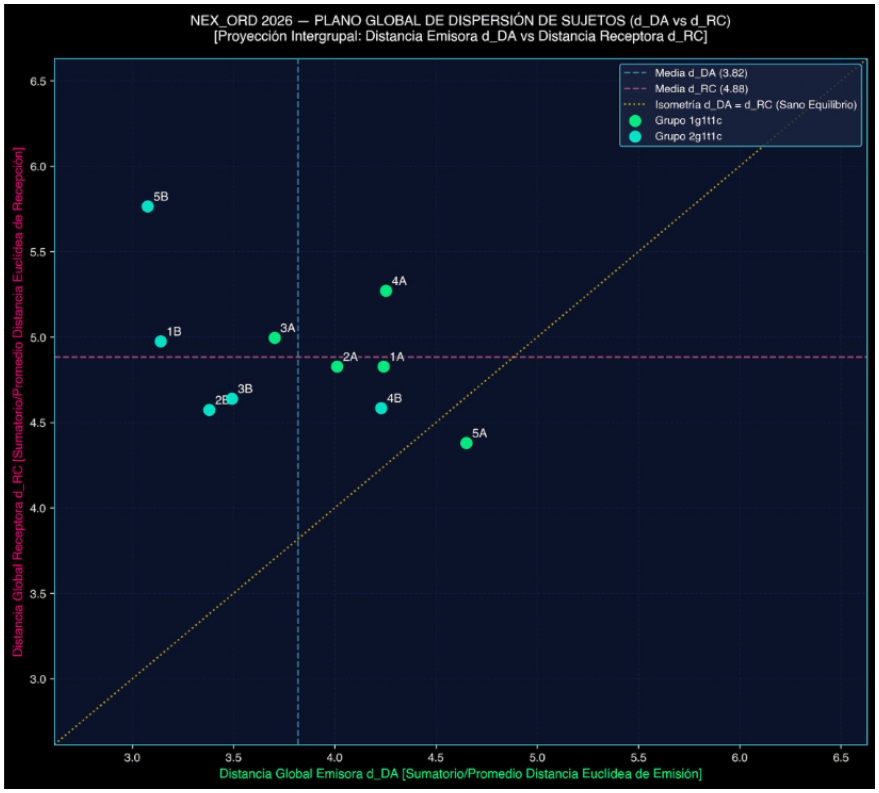


Figure 7. Scatter plot of interdistances among group members as senders and as receivers of preferences.

- Hydrodynamic Sonification: Converts densities and frictions into continuous tonal streams, enabling auditory monitoring of collective turbulence or calm without overloading the analyst's visual attention.

The capability to compute field densities for large collectives without loss of analytical continuity resolves a historical limitation of traditional sociometry, which was typically restricted to small groups (N < 30) due to classical sociogram saturation (Figure 8).

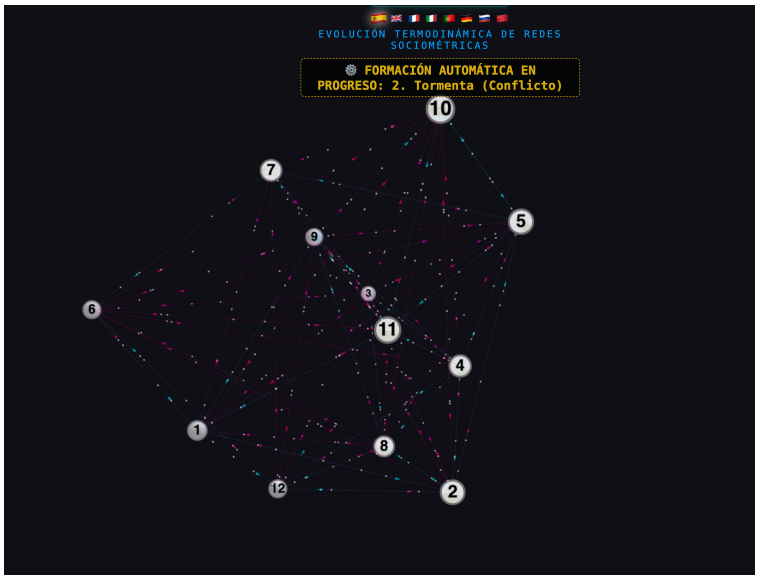


Figure 8. Interaction field and iceberg model in a collective of N = 50 subjects: macroscopic relational projection, attraction isolines, and structural ostracism detection.

6.4. The CULTURAL Corpus as a Zero-Reactivity Laboratory

The analysis of consolidated dramatic and literary masterpieces provides a first-order, non-reactive empirical baseline for psychosocial modeling. By transposing scripts and works into objective sociomatrices (e.g., Reginald Rose's *Twelve Angry Men*, Federico García Lorca's *The House of Bernarda Alba*, William Golding's *Lord of the Flies*), NEXORD analyzes complex group dynamics, turning points, and leadership crises in closed systems without introducing observer reactivity biases or relying on subjective observational inferences.

6.5. Triadic Social Cognition: Projected Expectations αE and ωE

NEXORD incorporates the triadic level of social cognition into the sociometry of dyadic metaperceptions and expectations. By asking participants to estimate who they believe will be the most preferred member (αE, projected cohesion anchor) and the least preferred member (ωE, projected vulnerability/exclusion anchor) of the collective, the platform activates two macro-structural sensors:

- Perceptual Consensus on Leadership (αE): Quantifies the concentration of perceived social centrality. A high agreement rate on αE reflects unified shared representation of leadership, whereas multi-polar dispersion indicates latent competition for group guidance.

- Perceptual Sensitivity to Exclusion (ωE): Identifies acute peripheralization before explicit behavioral ostracism emerges. Convergence on a single ωE node reveals collective stigmatization, whereas dispersed perceptions reflect low shared threat and higher group tolerance.

Crossing projected expectations (αE, ωE) with effective relational choices (DA, RC) enables early detection of cognitive blindness, structural denial, and collective self-deception, providing key diagnostic indicators for conflict prevention and early warning.

6.6. In Silico Sociometric Simulation: The PLATINUM Engine

The realization of in silico experimental research within the platform is orchestrated by the PLATINUM simulation engine. This engine generates complete, synthetic sociometric matrices parameterized by field laws, serving as a methodological testbed and pedagogical simulator.

The researcher defines the desired scenario within the G × T × C design space (groups, temporal waves, criteria with verbatim prompts, sample sizes, and context domain) via a standardized parameter file or guided interactive wizard, governing collective morphology through families of continuous parametric modulators:

- (a) Thermodynamic: Target valence density, star node saturation, convergence tolerance.
- (b) Kinetic Gravity and Assembly: Cohesion, metaperceptual transparency, intergroup bias, thermal noise, reciprocity, triangulation, and latent factions.
- (c) Longitudinal Temporal: Structural inertia, grievance decay, and exogenous rupture shocks.
- (d) Contextual: V10 and A16 batteries and the catalogue of 10 calibration domains α / γ (Table 8).

Reproducibility is absolute: pseudo-random seeds guarantee bit-for-bit regeneration of any corpus, and batch sizing enables Monte Carlo replications. Each simulation is identified by a standardized key (SIMUL-...) and encodes subjects with formal positional labels (1A1a...), preserving privacy by design. SIMUL sits alongside PROPIO, REAL, and CULTURAL in the catalogue of four operational data categories:

- REAL (Empirical Field Corpora): Data obtained from living human collectives (work organizations, classrooms, sports teams, or clinical units), gathered via web capture protocols under strict informed consent, privacy safeguards, and an exclusively non-punitive formative commitment (§6.7 and §7.1). This mode enables the simulation engine to estimate default values—when appropriate or necessary—for Contextual Variables (V10) and Group Attribution Adjectives (A16) from sociometric data and complementary information provided by the Principal Investigator when direct field collection was unfeasible.
- PROPIO (Researcher Corpora): Unique sociomatrices and designs contributed directly by users, educators, or consultants for particular research, teaching, or intervention in their own natural groups; enables the engine to process investigator-designed data and, if appropriate, complete missing data to ensure analytical continuity.

- CULTURAL (Zero-Reactivity Laboratories): Relational networks extracted from dramatic scripts, film screenplays, or canonical literary works (*Twelve Angry Men*, *The House of Bernarda Alba*, etc.), representing closed and exhaustively documented relational systems where group dynamics unfold with zero observer reactivity (§6.4); in this mode, the system verifies coding and completes required analytical matrices.
- SIMUL (In Silico Simulation): Algorithmic synthetic scenarios generated by parameterizing field laws (attraction, friction, reciprocity, clustering), utilized for instrument calibration, methodological development, and research training without human participant fatigue or exposure.

The operational circuit articulates four sequential links:

1 Scenario definition via interactive wizard.

2 Factual generation of raw matrices by the simulation engine.

3 Ingestion by the factorial orchestrator (255 indicators, SMIb matrices, CA/PCA projections, and Grassmann geometry).

4 3D/4D VISORD payload projection and synchronization to promotional web.

Simulated corpora constitute, moreover, the reference ground truth against which system indicators (IDA, Φ, ICE, and Pentagram densities) can be verified under a priori known parametric regimes, as well as the substrate for public interactive visualizers in the VISORD ecosystem.

6.7. Web-Assisted Field Studies

NEXORD enables the logistical execution—practical, rapid, personalized, and with high validity control—of sociometric response collection across comprehensive and complex field designs administered entirely via web. The Principal Investigator serves as the sole coordinator and ultimate responsible entity: designing configurations (simultaneous groups, longitudinal collection waves, criteria and prompts, member rosters, contact channels), scheduling connectivity windows, and issuing to each participant a randomized digital questionnaire (.html) completed in private from their own smartphone.

The infrastructure integrates logistical validity controls at every step: prior informed consent—and, for minors, explicit parental/guardian authorization—; exclusive photo or avatar rosters for peer identification alongside randomized name listings that neutralize order bias; response time tracking and follow-up reminders for delayed members; and, upon study closure, issuance of a private, non-transferable access token allowing each participant to view their personal report: general group findings, individual relational metrics, and constructive suggestions for attitude and task restructuring, in strict adherence to non-stigmatizing restitution rights (§7.3).

Collectively, the sociometric versatility that NEXORD places in researchers' hands is comprehensive—covering any configuration within the G × T × C space across all four data modes—providing rigorous insight into relational invariants that historically remained challenging to isolate.

6.8. Future Horizons: Toward a Broader Scientific Sociometry\The historical evolution of sociometry, from J. L. Moreno's pioneering works in the 1930s to the non-parametric formalization of NEXORD, has primarily addressed the intragroup level: the individual as an electing unit and the immediate group as interaction space.

However, major societal coexistence challenges in complex modern contexts—from interdepartmental friction in large corporations to territorial fractures and geopolitical alignments—demand a qualitative leap: treating the entire group as an active relational unit within an Intergroup Ordinal Sociometry (IOS). This does not represent dogmatic doctrine, but an open horizon toward a mature scientific sociometry capable of addressing multi-level relational complexity at scale.

Cognitive-Emotional and Intergroup Substrate

Inter-collective relationships are rarely decided exclusively on economic balances or explicit contractual agreements. They are grounded in a dense substrate of social representations, shared stereotypes, identity affinities, and mutual perceptions of threat or leadership.

Classical social psychology anticipated this territory in foundational investigations of immense significance:

- Muzafer Sherif's summer camp field experiments (Robbers Cave), which demonstrated the rapid emergence of mutual hostility between competing collectives and how superordinate goals alone succeed in restructuring the relational field.
- Leon Festinger's core contributions on social comparison processes, elucidating the genesis of ingroup favoritism and outgroup discrimination: the systemic drive of groups to compare themselves with peers to sustain positive identity, generating spontaneous preferential inclusion toward members and derogation toward outsiders.
- Formulations on minimal groups and social categorization (Tajfel, Turner), demonstrating that the mere arbitrary categorization into distinct groups is sufficient to trigger discriminatory allocation and identity bias even in the absence of material conflict.

Extending the mathematical formalism of NEXORD to intergroup relations—where groups themselves act as nodes in higher-order signed networks—constitutes the natural scientific horizon for the physics of collective interaction.

7. Institutional Governance, Ethics by Design, and Open Science

7.1. Ethics by Design and Strictly Non-Punitive Use

NEXORD embeds ethical safeguards into the algorithmic core of its design (Ethics by Design). The system explicitly prohibits the deployment of sociometry for punitive purposes, disciplinary proceedings, terminations, or negative personnel filtering. Any computation conducted on living human collectives (REAL category) mandates formal authentication, oversight, and ethical responsibility under a duly accredited institutional Principal Investigator (PI).

7.2. Open Science and Custody in the Sociometric Bank (BSOC)

In strict adherence to Open Science and FAIR data principles (Findable, Accessible, Interoperable, Reusable), the methodological and mathematical foundations of the platform are publicly accessible.

Field data are anonymized using standardized alphanumeric codes (1A1a ... 5A1a) and secured via immutable cryptographic timestamping with the SHA3-512 algorithm (Clélie Protocol), deposited under the academic custody of the Sociometric Bank (BSOC).

The BSOC originates from the imperative to give retrospective utility to vast historical archives of past sociometric studies: BSOC was created to preserve and valorize the empirical heritage generated across decades of psychosocial research, allowing sociomatrices from historical investigations to be integrated into retrospective studies, comparative analyses, and meta-analyses. Once donated under the standardized SMIb protocol—which unifies both MORENO and ORBIT formats—this historical archive becomes reusable scientific capital under FAIR principles, preserving the empirical investment of generations of researchers.

The NEXORD Platform materializes through three integrated pillars:

1 The computational Python suite implementing the formal apparatus (subject to continuous review and improvement, distributed under institutional accreditation per §7.4).

2 This Open Access Scientific Manifesto with registered DOI on Zenodo/CERN.

3 The promotional website hosted on GitHub Pages, featuring the VISORD 3D/4D animation suite—interactive educational resources operating as visualizers of simulated or real anonymized studies—organized across a radial navigation interface.

7.3. Participant Rights and Constructive Restitution

Each participant receives unique, temporary personal credentials to verify the appropriate processing of their own data, without access to individualized identifiable data of peers.

Re-identification of nodes is strictly restricted to the Principal Investigator in contexts of professional psychological support or formative mediation. All restitution of results is reflexive, pedagogical, and constructive, oriented toward collective well-being and healthy group development.

7.4. Dual-Use Risk Management and Ethical Accreditation

Sociometric data capture highly sensitive interpersonal dynamics and must never be instrumentalized for exclusionary practices or discriminatory ranking. NEXORD adopts security by design: while methodological and theoretical formulations are open access under the CC BY 4.0 license, operational access to research instruments on human participants requires institutional accreditation and formal researcher commitment to non-punitive, formative use.

It is important to clarify the exact scope of this declaration. This Manifesto is a programmatic scientific document—presenting theoretical, methodological, and pragmatic foundations—: neither the Manifesto, nor the computational suite, nor its algorithms are

subject to restrictive commercial patents; the Manifesto is disseminated openly under CC BY 4.0, and ethical safeguards regulate operational access to instruments involving human subjects without restricting the dissemination of scientific knowledge.

8. Formalized Technical Annexes

8.1. Annex I: VISORD_demo

Interactive Group Formation and Simulation Environment

Within the visualization and experimental modules of the NEXORD platform, the interactive VISORD_demo environment is engineered specifically for dynamic three-dimensional and four-dimensional (3D/4D) exploration of structural formation processes in small collectives ($N < 51$). Its mathematical architecture models the socio-thermodynamic field as a system of interdependent particles subjected to attractive and repulsive dyadic forces derived from the SMIb sociomatrix and positional invariants SDR and BDR.

The environment interactively simulates transitions through four consecutive temporal stages in group genesis:

1. Phase T_0 (Initial / Baseline State): Unpolarized configuration of initial dispersion, where dyadic densities remain latent and the field exhibits negligible structural tensions.

2. Phase T_1 (Storming / Differentiation): Activation of affinity and rejection vectors; preference divergence, manifestation of perceptual misalignments, and initial self-image tensions (IDA).

3. Phase T_2 (Norming / Subgroup Consolidation): Crystallization of mutually stable dyads, formation of compact sub-structures, reciprocity clusters, and progressive alignment of expectations.

4. Phase T_3 (Performing / Integrated Cohesion): Quasi-stationary state characterized by thermodynamic equilibrium distribution, where functional roles are fully established and exchange balances stabilize.

To enable controlled experimentation and heuristic testing in academic teaching and physical laboratories, the environment incorporates three precalibrated benchmark scenarios:

- Scenario $N = 20$ (Organizational Dynamics and Substructures): Designed to monitor the simultaneous emergence of differentiated subgroups, competing leaderships, and internal fault lines.
- Scenario $N = 12$ (Board / Intensive Deliberation): Calibrated to analyze executive committees or deliberative juries, where consensus pressure exerts stress on the relational network.
- Scenario $N = 8$ (Task Micro-Team): Focused on strict interdependence, high visibility of dyadic asymmetries, and rapid functional role polarization.

The system features a continuous temporal navigation mechanism (4D Moviola), enabling the researcher to rewind, pause, or accelerate the temporal flow of group

interaction to identify critical bifurcations, trust collapses, or exact structural stabilization points.

Within the 3D visual space, the engine computes in real time:

- Alpha Node (α): Primary attractor pole or gravitational leader, characterized by high dyadic attraction density and elevated receptive status (SRC).
- Omega Node (ω): Peripheral node situated in minimum tie potential zones, enabling preventive diagnosis of structural ostracism risk before open behavioral fractures occur.
- Real-Time Spectral and Thermodynamic Metrics: Simultaneous visualization of the Fiedler second Laplacian eigenvalue (λ_2, algebraic connectivity and graph cohesion), Heider cognitive frustration (λ_1), and thermodynamic balances of gross mass (BDR) and relative balance (SDR).

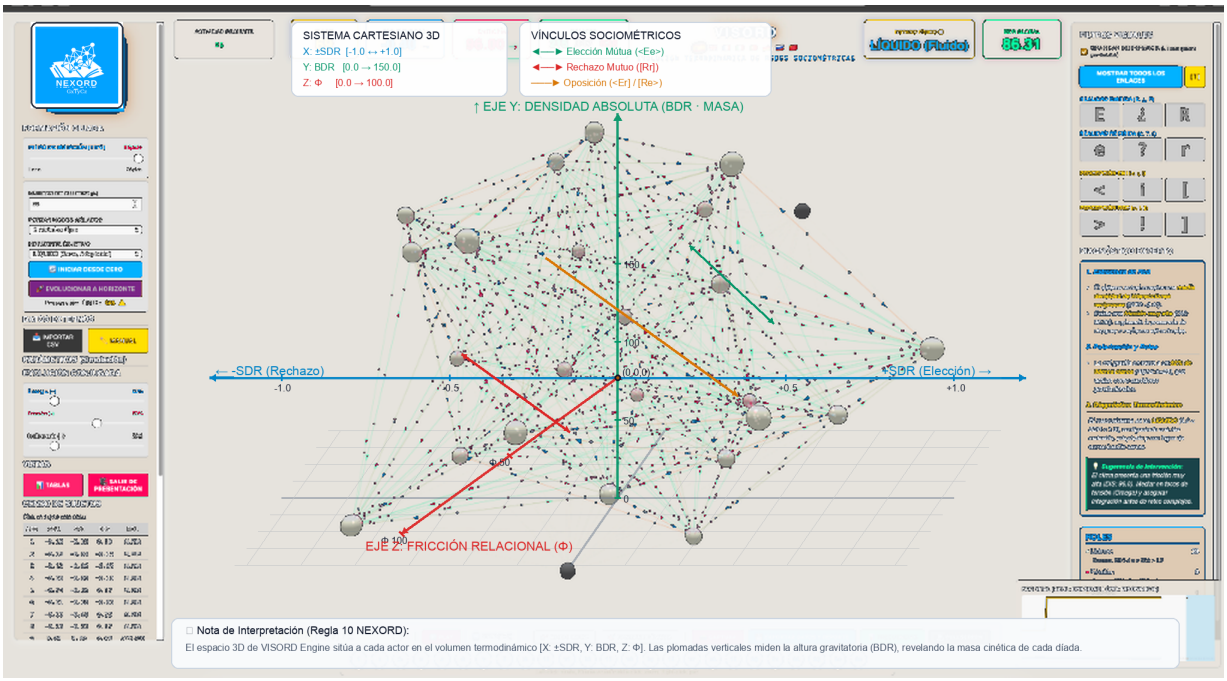


Figure Annex I. Interactive exploration environment VISORD_demo and 3D/4D visualization architecture for small groups (N < 51).

8.2. Annex II: Axonometric Projection via DIMO Algorithm (6 Factors in 2D)

The DIMO algorithm (Multidimensional Ordinal Informational Deployment) resolves the conventional projective limitation in factorial spaces when analyzing collectives exceeding twelve subjects (N > 12), in which relevant structural variance extends beyond the first two canonical factors. While habitual flat representations collapse or discard intermediate factors (third through sixth), the DIMO procedure integrates up to 6 simultaneous dimensions onto a single Cartesian canvas via an oblique orthogonal axonometric projection modulated by factorial inertia:

- Weighted Axonometric Coordinates: Each factor k (from 1 to 6) is projected along a regular radial distribution angle $\theta_k = (k - 1)\pi / K$, weighting its spatial contribution by the corresponding eigenvalue λ_k: $X_{DIMO} = \sum w_k \cdot f_k(i) \cdot \cos(\theta_k)$, $Y_{DIMO} = \sum w_k \cdot f_k(i) \cdot \sin(\theta_k)$.

- Vector Transition Traces: The algorithm draws crisp red vector transition lines connecting the basic two-dimensional anchor (axes 1 and 2) to the stabilized multivariate position in 6D space. The length and orientation of each trace quantifies dimensional tension and information gain: short trajectories denote actors whose dynamics are exhausted in primary factors; extended trajectories reveal actors whose alliances and tensions operate in higher-order dimensions.
- Latent Alliance Detection: By projecting all 6 factors onto the dark Cartesian canvas with attenuated KDE thermographic isobars, transversal alignments and sub-structures that remain invisible on conventional bifactorial planes become discernible.

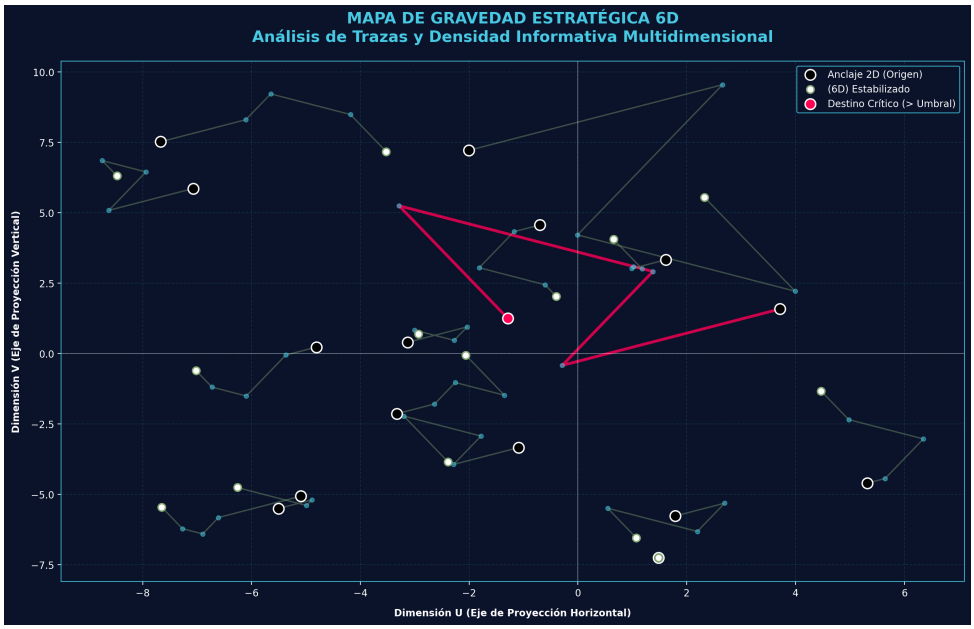


Figure Annex II. Axonometric Projection via DIMO Algorithm (6 Factors in 2D): Multidimensional Information Density Map with Crisp Red Vector Traces.

Pedagogical Note: The chart simultaneously deploys all six factors extracted from sociometric analysis. Crisp red traces indicate geometric displacement from the initial 2D baseline point to the corrected 6D location. Trace length measures information gain: short trajectories denote actors accounted for by conventional planes; long trajectories identify subjects with high relational complexity irreducible to two dimensions.

8.3. Annex III: 18D Relational Hyperspace and Dual Prism Synthesis

The 18D Relational Hyperspace formalizes the totality of interpersonal exchange balances through canonical decomposition of the four elemental positions of the SMIb ordinal tetragram (A_1 = p_DA, A_2 = DA, A_3 = RC, A_4 = p_RECEIVE). Its architecture structures the dyadic space across three complementary orthogonal prisms:

1. Prism 1 (Operational / Sender vs. Receiver Axis — 10D): Quantifies direct behavioral interaction through effective emission ($S_{12} = A_1+A_2$, $B_{12} = |A_1|+|A_2|$) and status reception ($S_{34} = A_3+A_4$, $B_{34} = |A_3|+|A_4|$), deducing role asymmetry $Diff_r = |B_{12} - B_{34}|$ and role energetic wear Diff_DR = BDR − SDR.

2. Prism 2 (Ontological / Mind vs. Facts Axis — 14D): Models cognitive coherence by decomposing the plane of crossed expectations ($S_{14} = A_1+A_4$, $B_{14} = |A_1|+|A_4|$) against binding

factual reality (S_{23} = A_2+A_3, B_{23} = |A_2|+|A_3|), identifying phantasmic inertia Diff_a = |B_{14} − B_{23}| and self-image discrepancy.

3. Prism 3 (Dialectical / Harmonic vs. Crossed Alignment Axis — 18D): Evaluates direct affective correspondence (S_{13} = A_1+A_3, B_{13} = |A_1|+|A_3|) versus crossed projected judgment (S_{24} = A_2+A_4, B_{24} = |A_2|+|A_4|), signaling dyadic disappointment risk Diff_i = |B_{13} − B_{24}|.

The entire system rigorously satisfies the Invariant Barycentric Theorem: for every actor and for the collective, Relative Density Balance fulfills strictly SDR = S_{12}+S_{34} = S_{14}+S_{23} = S_{13}+S_{24}, and Gross Relational Mass fulfills BDR = B_{12}+B_{34} = B_{14}+B_{23} = B_{13}+B_{24} (Δ = 0.000).

TABLA MAESTRA SÍNTESIS DENSIDAD RELACIONAL

PRISMA 1

					A1+A2	SRC-SDA	A3+A4	SDA+SRC	|A1|+|A2|	BRC-BDA	|A3|+|A4|	BDA+BRC	BDR-SDR
1g2t1c	A1	A2	A3	A4	SDA	Dif	SRC	SDR	BDA	Dif	BRC	BDR	Dif_DR
1	8,75	1,00	-0,51	8,64	9,75	-1,61	8,14	17,88	9,75	0,60	9,15	18,89	1,01
2	2,11	-2,22	-2,21	0,01	-0,12	-2,09	-2,20	-2,32	4,33	2,11	2,22	6,55	8,86
3	25,37	10,07	5,76	13,79	35,44	-15,90	19,54	54,98	35,44	15,90	19,54	54,98	0,00
4	-13,39	-15,84	-18,40	-16,82	-29,22	-6,00	-35,22	-64,44	29,22	-6,00	35,22	64,44	128,87
5	4,37	-1,62	-2,64	0,81	2,75	-4,58	-1,83	0,92	5,99	2,55	3,44	9,43	8,51
6	4,97	-3,03	-2,67	0,85	1,95	-3,77	-1,82	0,12	8,00	4,48	3,52	11,52	11,39
Media	5,36	-1,94	-3,44	1,21	3,42	-5,66	-2,23	1,19	15,45	3,27	12,18	27,63	26,44

PRISMA 2

					A1+A4	SRC-SDA	A2+A3	SDA+SRC	|A1|+|A4|	BRC-BDA	|A2|+|A3|	BDA+BRC	BDR-SDR
1g2t1c	A1	A2	A3	A4	SDA	Dif	SRC	SDR	BDA	Dif	BRC	BDR	Dif_DR
1	8,75	1,00	-0,51	8,64	17,39	-16,90	0,49	17,88	17,39	15,89	1,50	18,89	1,01
2	2,11	-2,22	-2,21	0,01	2,12	-6,55	-4,43	-2,32	2,12	-2,32	4,43	6,55	8,86
3	25,37	10,07	5,76	13,79	39,16	-23,34	15,82	54,98	39,16	23,34	15,82	54,98	0,00
4	-13,39	-15,84	-18,40	-16,82	-30,21	-4,03	-34,23	-64,44	30,21	-4,03	34,23	64,44	128,87
5	4,37	-1,62	-2,64	0,81	5,18	-9,43	-4,26	0,92	5,18	0,92	4,26	9,43	8,51
6	4,97	-3,03	-2,67	0,85	5,82	-11,52	-5,70	0,12	5,82	0,12	5,70	11,52	11,39
Media	5,36	-1,94	-3,44	1,21	6,58	-11,96	-5,38	1,19	16,64	5,66	10,99	27,63	26,44

PRISMA 3

					A1+A3	SRC-SDA	A2+A4	SDA+SRC	|A1|+|A3|	BRC-BDA	|A2|+|A4|	BDA+BRC	BDR-SDR
1g2t1c	A1	A2	A3	A4	SDA	Dif	SRC	SDR	BDA	Dif	BRC	BDR	Dif_DR
1	8,75	1,00	-0,51	8,64	8,25	1,39	9,64	17,88	9,26	-0,38	9,64	18,89	1,01
2	2,11	-2,22	-2,21	0,01	-0,11	-2,11	-2,21	-2,32	4,32	2,09	2,23	6,55	8,86
3	25,37	10,07	5,76	13,79	31,13	-7,28	23,85	54,98	31,13	7,28	23,85	54,98	0,00
4	-13,39	-15,84	-18,40	-16,82	-31,78	-0,88	-32,66	-64,44	31,78	-0,88	32,66	64,44	128,87
5	4,37	-1,62	-2,64	0,81	1,74	-2,55	-0,82	0,92	7,01	4,58	2,43	9,43	8,51
6	4,97	-3,03	-2,67	0,85	2,30	-4,48	-2,18	0,12	7,64	3,77	3,88	11,52	11,39
Media	5,36	-1,94	-3,44	1,21	1,92	-2,65	-0,73	1,19	15,19	2,74	12,45	27,63	26,44

Table Annex III. Specification and Calculation of the 18 Relational Densities and Socio-Thermodynamic Invariants (Operational Prism 12-34, Ontological 14-23, Dialectical 13-24, and Global Balances SDR / BDR).

Pedagogical Note: Notice in the empirical table how all three prismatic decompositions yield strictly identical values for net relative balance (SDR = 17.88, −2.32, 54.98, −64.44, 0.92, 0.12; Mean = 1.19) and gross mass (BDR = 18.89, 6.55, 54.98, 64.44, 9.43, 11.52; Mean = 27.63 across ontological and dialectical prisms), irrefutably confirming barycentric invariance. The Diff_DR column quantifies the internal dissipative inertia of the group.

8.4. Annex IV: Spectral Modeling (FFT) and Gravitational Graph (Markov) — Comparative Analysis of Figures

Beyond static sociometric description, the NEXORD platform incorporates two advanced procedures from relational physics: spatial harmonic frequency decomposition via Fast Fourier Transform (FFT on AGE Grassmann manifolds) and stochastic energy transfer modeling via Markov

Chains and Gravitational Graphs (N7_PREDICTIVE module).

A. FFT Spectral Decomposition and Tie Friction

When relational senders are ordered along their decreasing structural status gradient (from central leaders to isolated peripheral nodes), the distribution of assigned ordinal ranks describes a characteristic discrete spatial signal. The FFT decomposes this signal into two fundamental orthogonal components:

1. The Fundamental Structural Trend (Frequencies F_0 and F_1): Represents the coherent baseline, social role stability, and low-perturbation preferences.

2. The Raw Noise / Friction Wave (High Frequency F_high): Captures abrupt oscillations, affective ambivalence, and relational entropy, enabling calculation of the Spectral Friction Index (SFI / IFE): **SFI = [∑_k > 1 |A(k)|^2 / ∑_k ≥ 0 |A(k)|^2] · 100**.

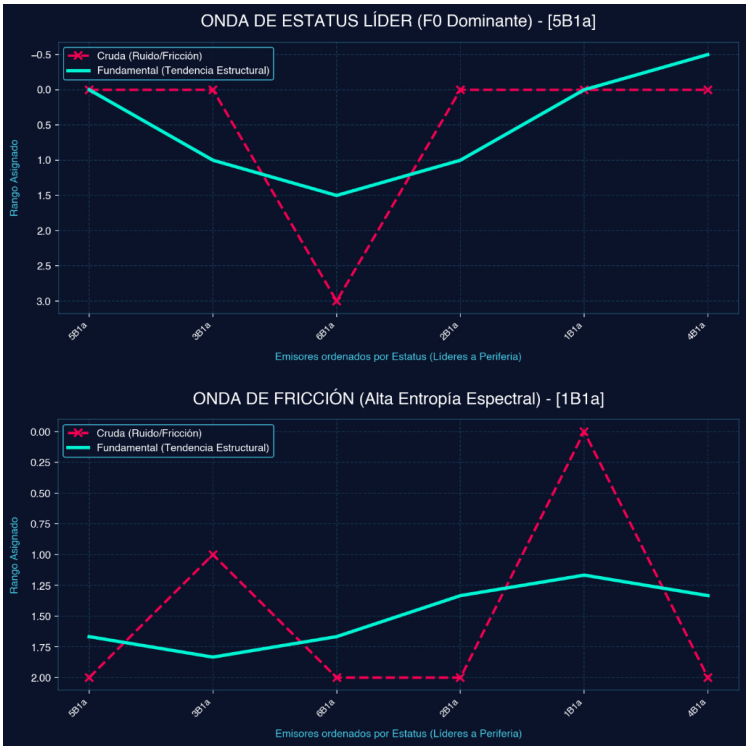


Figure Annex IV-A. FFT Spectral Decomposition of Relational Waves: Leader Status Wave (5B1a, Dominant Frequency F_0) versus Tie Friction Wave (1B1a, High Spectral Entropy).

Analysis of Figure Annex IV-A: The upper panel illustrates the dynamics of a consolidated status leader (5B1a). The fundamental trend (cyan solid line) coincides harmoniously with raw choices (magenta dashed line with markers), denoting minimal friction (low SFI), high group consensus, and absence of disruptive noise. In sharp contrast, the lower panel depicts an actor in acute friction (1B1a): raw data oscillate in sharp sawteeth (jumping abruptly between extreme ranks 0.00 and 2.00), diverging strongly from the structural trend. This spectral divergence reveals ambivalence, tie polarization, and high internal thermal dissipation.

B. Markov Chains and Stochastic Gravitational Graph

The Markovian model formalizes a dynamic system where choice energy flows through dyadic channels weighted by the physical law W(k) = 2^(1−k). The normalized transition matrix P is regulated via an ergodic teleportation parameter α = 0.85 (damping factor): M = αP + ((1−α)/N) 1 1^T. The stationary eigenvector π associated with the dominant eigenvalue (πM = π) formalizes the Sociometric Gravitational Mass (%):

- Recursive Gravity versus Flat Tally: Unlike classical sociometric indices that merely sum direct nominations as if all votes carried identical weight, the Markovian procedure evaluates who emits the nomination. Receiving the preference of a gravitationally massive actor transfers exponentially greater energy than multiple nominations from the periphery.
- Asymptotic State Prediction: By multiplying the positional distribution at T_1 by successive powers of the stochastic matrix P^t, the system predicts the asymptotic probabilistic evolution of the collective at infinite time (T_∞), identifying integration attractor states or absorbing exclusion regions.

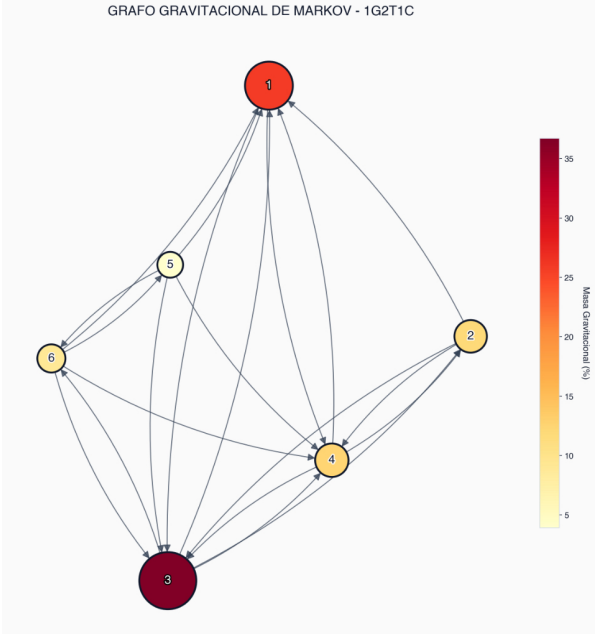


Figure Annex IV-B. Markov Gravitational Graph (1G2T1C): Directed Stochastic Flow Network and Stationary Gravitational Mass Distribution (π).

Analysis of Figure Annex IV-B: In the directed graph, node diameter and thermal shading (graduated from pale yellow to deep garnet red along the sidebar) represent the Stationary Gravitational Mass (%) of each subject. Actor 3 emerges as the dominant gravitational sink of the collective, retaining approximately 36% of the system's total energetic mass, followed by Actor 1 (~28%). Conversely, Actors 5, 6, and 2 appear as peripheral satellites of minimal mass (4% to 8%), channeling their flow toward the central 1–3 axis without achieving autonomous retention.

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3 Ye, K., & Lim, L.-H. (2016). Schubert varieties and distances on Grassmann manifolds. SIAM Journal on Matrix Analysis and Applications, 37(3), 1176–1197. https://doi.org/10.1137/15M1047648

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